



East West University

Task 1 Report

Course Code: CSE475

Course Name: Machine Learning

Section: 2

Group	Group04
Dataset	BloodMNIST (MedMNIST v2)
Track	Track 2 + Track 3 (GCN + CNN+Att)
Proposed Direction	CNN + Attention + GCN

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CSE475 Machine Learning - Task 1 Report (EDA and Related Work)

1. Introduction

This report summarizes the Task 1 exploratory data analysis and related-work review for BloodMNIST. BloodMNIST is a lightweight biomedical image classification dataset from MedMNIST v2. The goal of the project is to classify blood-cell microscopy images into eight categories. Our selected modeling direction is a hybrid CNN + Attention + GCN architecture, where a CNN first extracts local image features, an attention module emphasizes informative regions or channels, and a graph convolutional network learns spatial relationships between patch-level image regions. This task focuses on understanding what the data looks like, checking data quality, studying class balance, reviewing relevant papers, and identifying a research gap that motivates the proposed model.

2. Dataset

BloodMNIST is an 8-class blood-cell image classification dataset from MedMNIST v2. The images are RGB microscopy images resized to 28 x 28 pixels. The dataset is already provided with official train, validation, and test splits with 11,959 train, 1,712 validation, 3,421 test images. It has eight classes and they are Basophil, Eosinophil, Erythroblast, Immature Granulocytes, Lymphocyte, Monocyte, Neutrophil, and Platelet.

Table: Dataset Overview

Item	Finding
Dataset	BloodMNIST from MedMNIST v2
Problem	8-class blood-cell image classification
Samples	17,092 total: 11,959 train, 1,712 validation, 3,421 test
Image format	28 x 28 RGB images, uint8, pixel range 0-255
Classes	Basophil, Eosinophil, Erythroblast, Immature Granulocytes, Lymphocyte, Monocyte, Neutrophil, Platelet
Class balance	Moderately imbalanced; training majority/minority ratio about 2.74:1
Data quality	No missing values, invalid labels, infinite values, or blank images found
Duplicate audit	Exact duplicates were reported, including possible cross-split duplicates; official split preserved for fair benchmark comparison

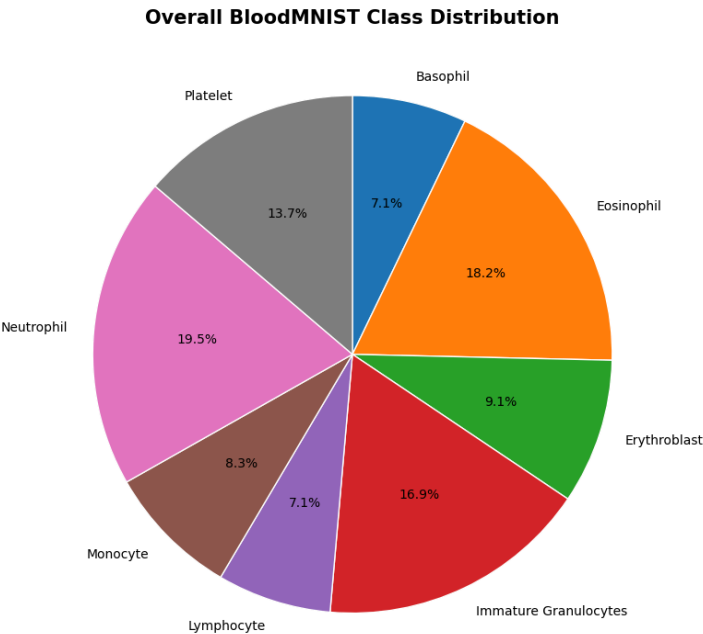
3. EDA Findings

Class Imbalance Summary

No.	Metric	Value
0	Majority Class	Neutrophil
1	Minority Class	Lymphocyte
2	Majority Images	3329
3	Minority Images	1214
4	Imbalance Ratio	2.74:1
5	Dataset Status	Moderately Imbalanced

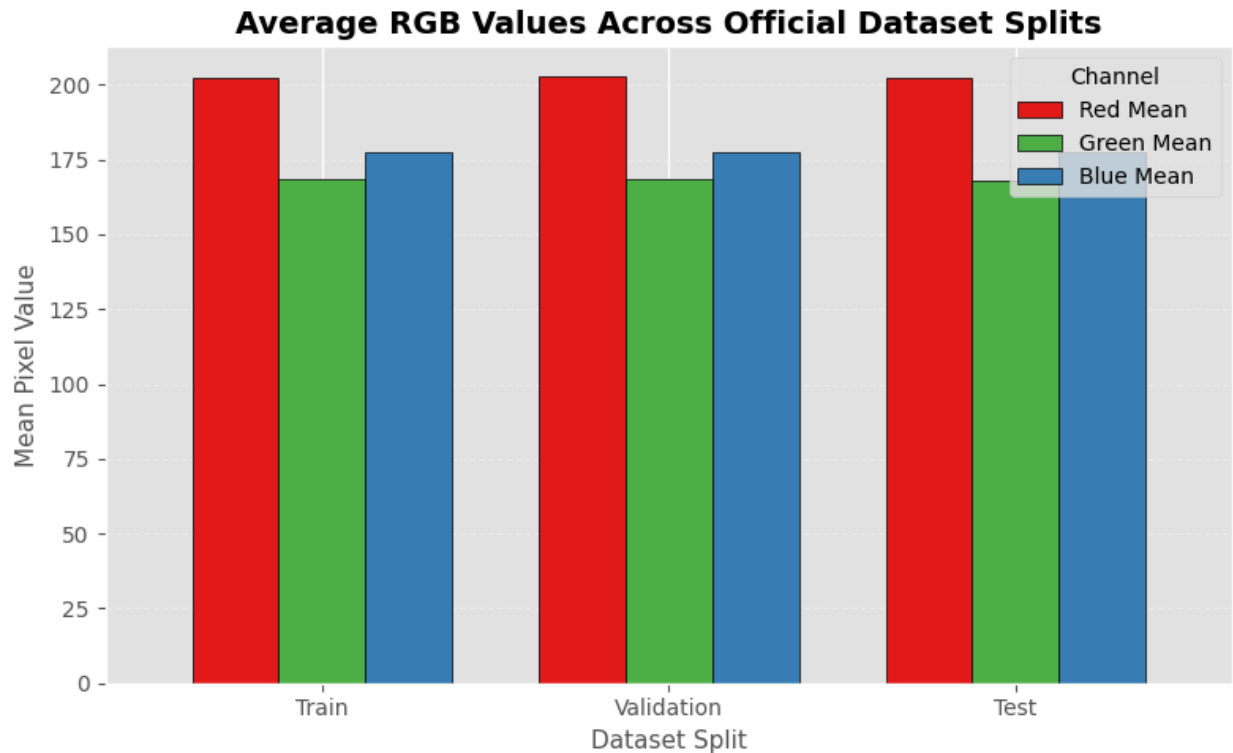
The dataset shows moderate class imbalance, where Neutrophil is the majority class and Lymphocyte is the minority class with an imbalance ratio of approximately 2.74:1. Therefore, accuracy alone may not fully represent model performance.

Output of Cell-6A:



This figure provides an intuitive overview of the proportion of each blood-cell class in the complete BloodMNIST dataset. It helps identify whether any class dominates the dataset and supports the imbalance analysis performed in the previous section.

Output of Cell-9A:



This section compares the average Red, Green, and Blue (RGB) channel intensities across the official Training, Validation, and Test splits.

The objective is to verify whether the three official dataset splits have similar colour distributions. Consistent RGB statistics indicate that the official MedMNIST split is balanced with respect to global colour characteristics and does not introduce obvious colour bias between training and evaluation data.

Brightness		Contrast		Blur Score		Entropy		Edge Strength		Center-Border Gap		
	mean	std	mean	std	mean	std	mean	std	mean	std	mean	std
Split												
Test	179.4 68994	15.38	52.42 1001	9.929	1529. 404	557.7 66	6.582	0.429	13.028	2.193	88.345	33.661
Train	179.5 44006	15.439	52.35 2001	9.904	1518. 534	550.9 99	6.577	0.434	12.984	2.226	88.216	33.57
Validation	179.5 85007	14.989	52.21 0999	9.802	1530. 34	558.0 36	6.591	0.424	13.053	2.233	88.473	33.252

This section summarizes the image quality metrics across the official Training, Validation, and Test splits. The objective is to verify that the dataset has consistent image characteristics before model development. The reported values are descriptive statistics only and are **not** used for filtering or modifying the official BloodMNIST dataset.

Output of 12A:



Only a small proportion of images fall into the extreme ends of the training-set quality distributions. These samples are flagged as **potential quality outliers** and require visual

inspection before any decision regarding removal. The official MedMNIST dataset splits remain unchanged throughout Task 1 to ensure fair benchmarking and reproducibility.

Output of 14:

Within-Split Duplicate Summary

No	Split	Total Images	Unique Images	Extra Duplicate Images	Duplicate Groups	Duplicate %
0	Train	11959	11951	8	8	0.067
1	Validation	1712	1712	0	0	0
2	Test	3421	3421	0	0	0

4. Related Work Summary

Table: Related Work Summary

Paper	Method	Key Result	Gap / Relation
MedNet, Ferdous et al. (2025)	Lightweight CNN with CBAM attention	BloodMNIST AUC 0.998, accuracy 0.983	Strong attention-CNN baseline, but no graph conversion
Duc et al. (2025)	YOLO-v10 + CentreNet + SSO + multi-hop attention GNN	WBC accuracy 99.18%	Graph method is complex and not tested on low-resolution BloodMNIST
Zhao et al. (2026)	Fuzzy attention CNN	BloodMNIST macro-F1 about 0.96	Models uncertainty but does not use GCN-based image graphs
Altarawneh (2023)	Hybrid CNN + GNN for leukocyte detection	Accuracy 96.87%	Relevant hybrid idea, but graph details and GNN type are unclear
Halder et al. (2024)	Vision Transformer on MedMNIST v2	BloodMNIST accuracy 97.90%	Uses self-attention, but no CNN-GCN graph learning

5. Research Gap

Recent studies have shown that CNN-based models, attention mechanisms, and GNN-based approaches can effectively improve medical image classification. CNNs are widely used for extracting local morphological features from blood cell images, while attention mechanisms enhance important feature regions. However, CNN-based approaches mainly focus on local feature extraction and have limited ability to capture spatial relationships between different regions of a cell.

Hybrid CNN-GNN approaches have attempted to address this limitation by introducing graph-based relational learning. However, existing methods often lack an effective feature refinement mechanism before graph construction or rely on relatively simple graph processing strategies. On the other hand, transformer-based approaches can capture long-range dependencies but may lack the local inductive bias required for fine-grained medical image analysis.

For BloodMNIST classification, distinguishing between visually similar blood cell categories requires both detailed local morphology and meaningful relationships among different cell regions. Existing approaches usually address these aspects separately rather than combining them in a unified framework.

Therefore, there is a need for an integrated architecture that combines local feature extraction, attention-based feature enhancement, and graph-based relational reasoning. This project addresses this gap by proposing a **CNN → Attention Module → Graph Construction → GCN** framework. The CNN extracts local visual features, the attention module highlights informative regions, and the GCN learns spatial relationships from the generated feature graph. This combination aims to provide a more comprehensive representation of blood cell morphology for BloodMNIST classification.

6. Conclusion

BloodMNIST is a clean, compact, and suitable dataset for Task 2 model development. The EDA confirms that the dataset has valid images, consistent official splits, moderate class imbalance, and visually overlapping blood-cell classes. These findings motivate a model that can learn local morphology, focus on important regions, and test whether spatial graph relationships add value beyond CNN and attention features.

References

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